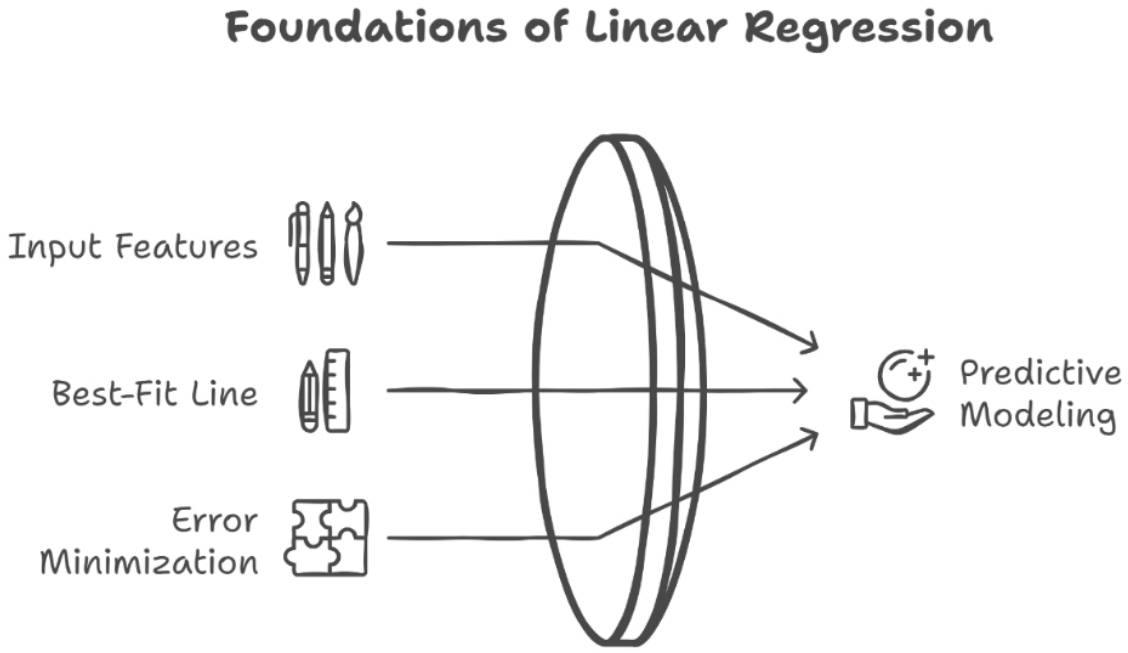
values. This algorithm is particularly effective for understanding and predicting continuous outcomes based on input features.



Why Use Linear Regression?

Linear regression is widely used because it is:

- **Easy to understand and interpret:** The coefficients of the regression equation directly indicate the relationship between input variables and the target.
- **Computationally efficient:** Linear regression is computationally lightweight, making it suitable for large datasets.
- **Versatile:** It can be extended to multiple linear regression to handle multiple input features.

However, it is important to note that linear regression assumes a linear relationship between the input features and the target variable. If this assumption does not hold, the model's predictions may not be reliable.